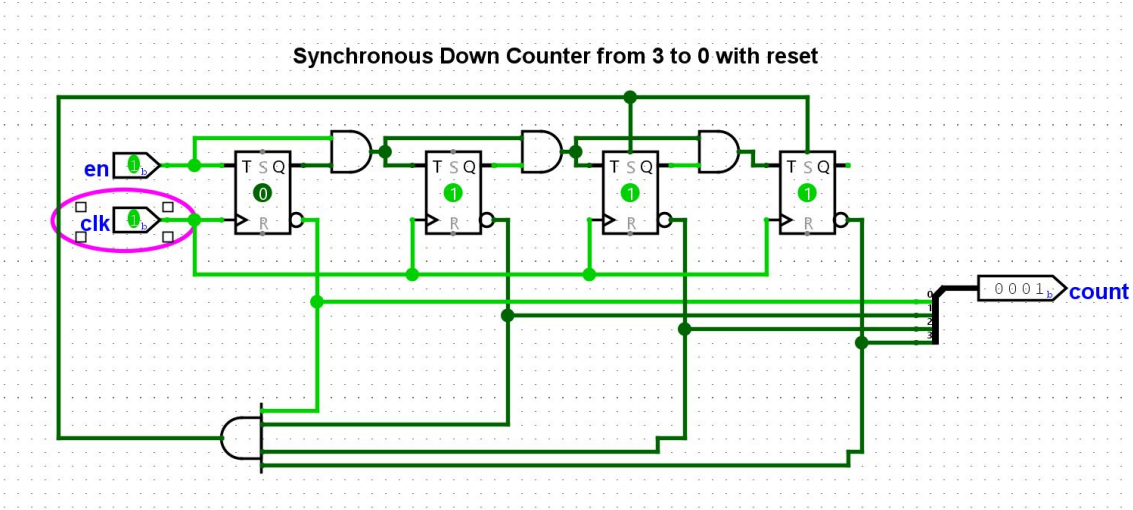
Lab 07

Chelsea Sisneros

Fixing the count down from 10-0 to 3-0.



Right now this image is stopped at the number 3, after that it will go to 2, 1, 0, then reset.

Finite State Machine Designs

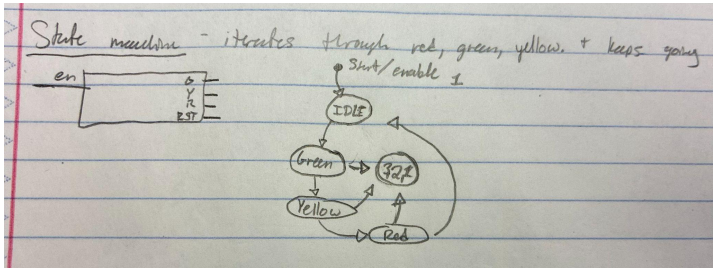


Image 1

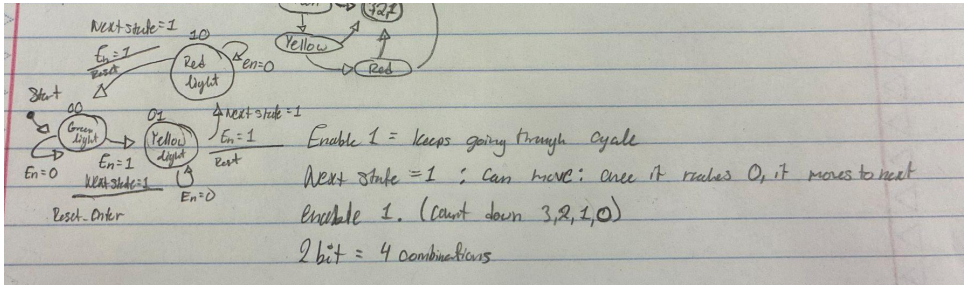


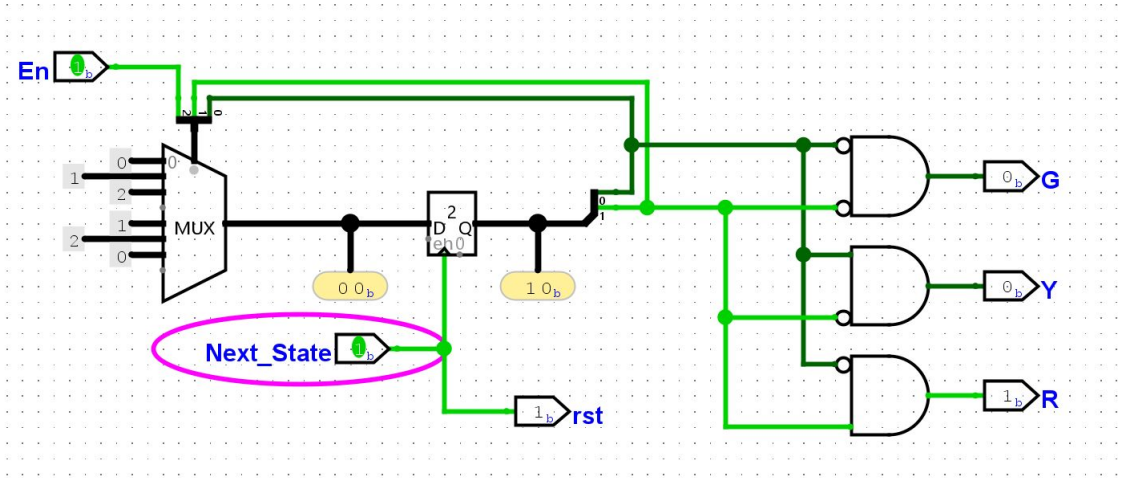
Image 2

Image 1 has the FSM I created and Image 2 has the FSM that should work properly.

My FSM was missing the enable, reset, and next state for each light. On mine I thought the lights all got connected to a timer and after that they should be good to go. But, you need the next step so that the machine understands when the timer goes back to 0 and can move on to the next color.

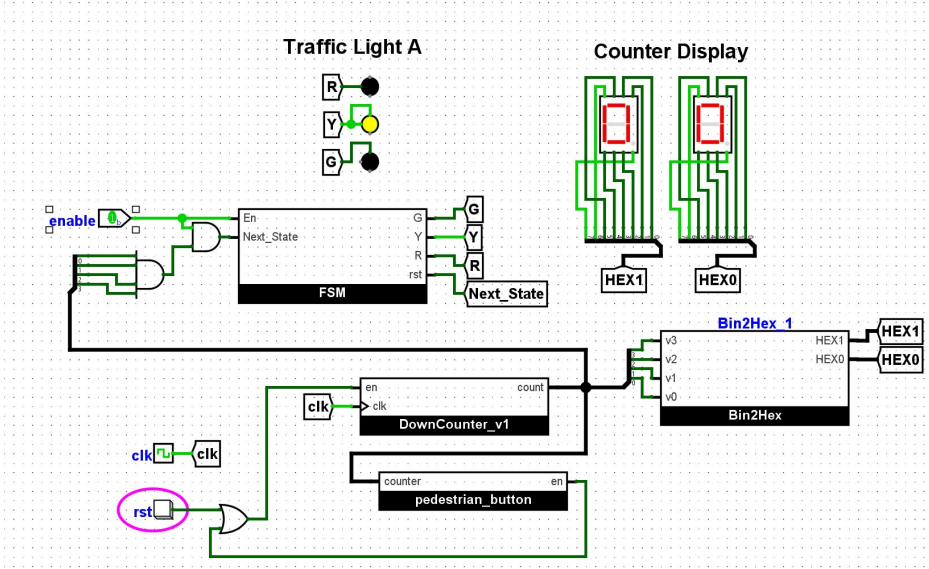
Output = Reset (Reset_Counter)
G = 1
Y = 1
R = 1

Finite State Machine Designs



This is the Final FSM machine built in Logisim. After you turn on the enable and the rst (reset) button, you can click the Next State button and move through green, yellow, and red while each output color waits for the timer to countdown from 2-0.

Connecting Everything Together



Here we connected the FSM to the rest of the logic. By connecting the FSM we are able to see the color of our traffic lights changing (green, yellow, red), this specific image is showing yellow! The Counter Display counts down from 0-3 once the reset button gets pushed. It then signals the FSM to wait the 3 seconds before changing to the next color (this is from the Next_State output). It moves to a different color once the reset button gets pushed.